ACA Assignment - 3

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Analyze features of ARM processors in designing applications

1] Features of ARM debug architecture

- Breakpoint :- Stops program execution at a specific instruction for debugging
- Watchpoint :- Monitors specific memory locations and stops execution when they are accessed or modified
- Debug Access port :- Allows an external debugger to access the processor and its memory
- JTAG / Wire :- provides a connection b/w the processor and external debugging tool
- Single stepping :- Executes one instruction at a time to check program behaviour
- Debug register :- Store control and status information reqd for debugging.
- Trace Support :- Records program execution to help identify errors and analyse performance

2] Analyse ARM TDM1 3 stage pipeline :-

-> ARM TDM1 uses a 3 stage pipeline to improve instruction execution speed. It allows multiple instructions to be processed simultaneously.

The three stages are :-

- Fetch :- the processor fetches the instruction from memory
- Decode :- the instruction is decoded and required operation is identified

Execute :- the processor performs the operation and stores the result.

working

- while one instruction is being executed the next instruction is decoded
- At the same time another instruction is fetched
- This overlapping increases processor performance

3] Analyse the necessity of memory system

A memory system is required in a computer to store instructions, data and results so that the CPU can access them whenever needed

Importance of memory system :-

- Data storage
- Fast access
- CPU Performance
- Temporary Storage
- Permanent storage
- Efficient processing
- Data transfers

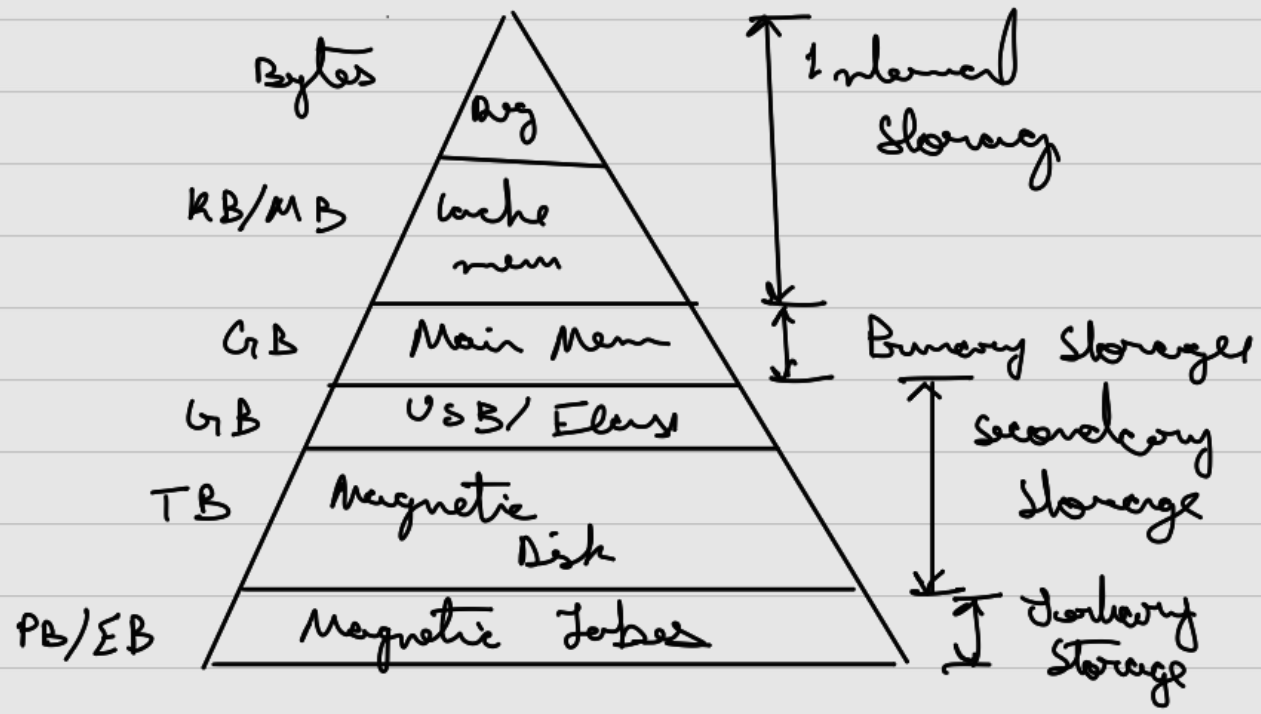
4] Analyze and explain memory hierarchy with necessary diagram.

→ Memory Hierarchy

Memory hierarchy is the arrangement of different types of memory in a computer system based on speed, cost and storage capacity.

It is designed to provide faster access to frequently used data while making large storage capacity at a reasonable cost.

Memory Hierarchy Diagram :-



levels of memory hierarchy :-

1. Registers
2. Cache memory
3. Main memory
4. Secondary memory
5. Analyse and explain memory mangemt systems

→ Memory Mangemt system

Memory mangemt is the process of controlling and organising the computers main memory (RAM). It manages the allocatin and deallocatin of memory to different prog during executin.

Functn of memory mangemt :-

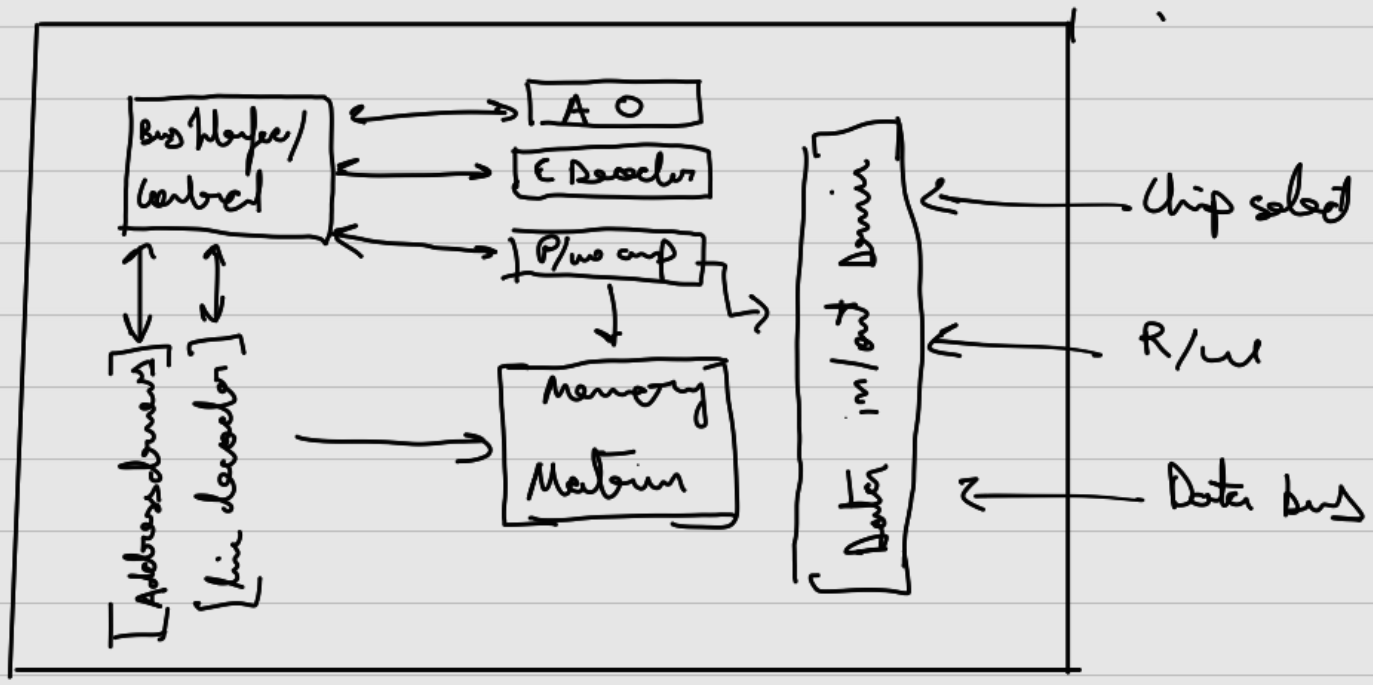
1. Memory allocatin
2. Memory de-allocatin
3. Memory protectn
4. Address translatin
5. Memory sharing
6. Memory utilizatn

Types of memory mangemt

- Contiguous allocatn
- Segmentatn
- Paging
- Virtual memory

6] Analyse and explain block diagram of RAM chip and its operatn with CPU.

-> Block Diagram of RAM chip and it operate with CPU



Main Blocks

- Address Bus ÷ Carries the memory address from CPU
- Address Decoder ÷ Selects required memory location
- Memory Array ÷ Stores bits in binary form
- Data bus ÷ Transfers data b/w RAM & CPU
- Read / write ÷ Controls read and write operation
- Chip select ÷ Enables the RAM chip

Read operation :-

1. CPU sends the memory address
2. Address decoder selects the location
3. Read signal is activated
4. Data is transferred from RAM to CPU

Write operation :-

1. CPU sends the memory address and data
2. Address decoder selects the location
3. Write signal is activated
4. Data is stored in RAM

7) Analyze and explain various technology used in RAM

-> Various tech in RAM

RAM is mainly classified into two types based on the technology.

SRAM

- Stores data using flip flops
- Does not require refreshing
- Very fast but expensive
- Used in cache

DRAM

- Stores data using capacitors and transistors
- Requires periodic refreshing
- Slower than SRAM but cheaper
- Used as main memory (RAM)

Types of DRAM

- SDRAM
- DDR, RAM
- DDR2, DDR3, DDR4, DDR5

2] Analyze and explain types of RAM chips and its operation with CPU

→ ROM (read only memory] is a non-volatile memory used to store permanent data and instructions. It retains data even when power is off

Types of ROM

PROM (Programmable ROM)

- Can be programmed only once
- Data cannot be erased

EEPROM :-

- Erased and re programmed electrically
- Data can be updated

Flash ROM

- A type of EEPROM
- Erases and works data in blocks
- Used in USB drives and memory cards

operation of ROM with CPU

- CPU sends the required address through the address bus
- Address decoder selects the corresponding memory location
- Stored data is accessed from the memory array
- Data is transferred through the o/p buffer
- CPU receives the data through the data bus

9] Auxiliary and explain auxiliary and cache memory

→ Auxiliary memory is a non volatile secondary storage is used to store large amounts of data and keep permanently.

Features :-

1. Stores data permanently even when power is off
2. Has large storage capacity
3. Slower than main memory
4. Cheaper than primary memory
   Ex → HDD, SSD, CD

Operation

- CPU requests data from secondary storage
- Required data is transferred to RAM
- CPU access the data from RAM for processing

Cache Memory

Cache memory is a small high speed, memory located inside or near the CPU. It stores freq. used data & instructions.

Features :-

1. Very fast memory
2. Smaller storage capacity
3. More expensive than RAM
4. Reduces CPU memory access time
5. Store frequently accessed data

Levels of Cache

L1: Fastest & smallest cache
L2: Larger than L1 but slower
L3: Larger than L2 and shared b/w CPU cores in many processors

operations

- CPU request data
- Cache check whether data is available
- If found data is sent to CPU
- If not found data is fetched from RAM and stored in cache

10] Analyze and explain temporal locality

→ Temporal and spatial locality, locality of ref means the CPU tries to access the same or nearby memory location repeatedly.

Temporal locality

Recently accessed data is likely to be accessed again.

Ex:- A variable used repeatedly in a loop

Spatial locality

Nearby memory locations are likely to accessed soon

- Ex :- Accessing array elements one after the other.

11) Analyze and explain mapping functions and techniques with respect to memory make short

→ Mapping function determines how block of main memory are placed in cache memory

- Direct mapping
- Each memory block maps to only one fixed cache line
- Simple and fast
- May cause more conflicts

Associative Mapping

- Any memory block can be placed in cache line
- Flexible but expensive
- Requires searching multiple cache lines

- Set Associative Mapping

- Cache is divided into sets
- Memory block maps to a particular set but can occupy any line within that set.
- Combines advantages of direct and associative mapping

12] Analyse and explain virtual memory, paging, page replacement, page fault, segmented page map, TBL and memory protection

Virtual memory

- Allows prog larger than RAM to execute
- Use secondary storage as an extension of RAM

Paging

- Divides memory into fixed size blocks
- Virtual blocks are called pages and physical blocks are called frames.

Page Replacement -

- Replaces an existing page when RAM is full

Algorithms :- FIFO, LRU & optimal

Page Fault

- Occurs when required page is not in RAM
- OS loads the page from secondary storage

Segmented Page Map

- Combines seg & paging
- Divides program into segments and then into pages.

TLB

- Small high speed memory in the MMU
- Stores recently used page table entries
- Speeds up address search

Memory Protection

- Prevents unauthorised memory access
- Protects one process from another.

13] Analyse and explain operation of nano memory with example

-> Refers to memory technology that uses nanoscale components to store and access data.

operation

- Uses very small electronic components to store binary data
- Data is stored by changing the electrical state or reinforcement of memory cells
- The memory controller reads and writes data
- Stored data can be accessed when required.

Ex -> NAND flash memory

Advantages :-

- Small size
- High storage capacity
- Low power consumption

14) Consider length of instruction field 12 bits and address field is 4 bit each. There is two address, zero one address operation. Find out number of 0 address operation.

-> Instruction field = 12 bit
Address field = 4 bits
Two address instruction = 2^4

Find zero address instruction

Four address instructions

Each instruction uses 2 address fields

- Remaining instruction bits

$$12 - 4 - 4 = 4$$

Number of available combinations

$$2^4 = 16$$

15 are used

$$16 - 15 = 1$$

Step 2 one address instructions ÷

$$1 \times 2^4 = 16$$

4 one address instruction are given

$$(16 - 1) \times 16 = 128$$

Step 3 ÷ zero address instruction

$$128 \times 2^4 = 2048$$

∴ 2048 zero address instructions